**Machine vision models, how to choose, train, and integrate one into a system**

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ABSTRACT

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LIST OF ABBREVIATIONS

| | |
|---|---|
| AI | Artificial Intelligence |

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Abstract

Tiivistelmä

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# 1 Introduction

Machine vision is a tool to make computer understand images (Komeili, 2023) and this master's thesis will attempt to create a machine vision model while documenting the whole process. This is a research done for Bitwell (*Bitwell Oy*, no date) and this research is go over what machine vision models are available, how to train them, what camera systems can be used, compare options and optimize cost, and produce an artifact which is a working machine vision system for Bitwell. Thus this research paper will answer to questions:

1. How to implement a taught machine vision model in real time image manipulation and analyzes?

2. Which available machine vision model is most optimal in terms of cost while usable in general situations?

3. What tools and requirements are there to taught a machine vision model?

4. What services and equipment are available to taught a machine vision model and how to choose least costing one while requirements are met?

5. What interfaces are there and how to utilize them in user applications?

6. What software development tools are there?

7. Can machine vision model be run on a embedded system? What options and limitations are there?

The importance of this research comes from implementation of improved machine vision and guide how to choose what tools to use to implement this improved machine vision.

(TODO: most significant results and why they are significant)

The structure of this document will be following. The first chapter is the ”Introduction” where issues and research questions are presented. The second chapter is the ”Related research” where foundation and comparison points for this research are presented. The third chapter is the ”Methodology” where the used research methodologies are explained in detail and how this research utilizes them. The fourth chapter is the ”Machine vision models” chapter that explains how machine visions work, compares different models, and explains

why this research chose a model to use. The fifth chapter is the ”Training of the machine vision model” which explains the training process used to train machine visions and how this research did it. The sixth chapter is the “camera systems” which compares different camera systems and tells how to choose correct one. The seventh chapter is the “the artifact” where the implementation is documented in the detail and how decisions affected the outcome. The eighth chapter is the “Discussion” where results are analyzed and compared to “Related research” chapter. The ninth chapter is “Conclusions” where main questions and results are summarized. The tenth chapter is “Future work” where questions to next research are presented.

# 2 Related research

This chapter will present similar studies and research as a foundation to this research. This is as comparison to show what new knowledge this research will bring.

## 2.1 Machine vision

This sub-chapter will present how other have implemented their machine vision model and how that compares to this research.

A book called “Machine learning” defines computer vision as a discipline inspired by the human visual system (Theodoridis, 2026, p. 7). (TODO: add more definitions). Alas, the used definition of machine vision here will be system that analyses data from cameras.

Halinen’s master’s thesis research focuses on theory on what requirements does a machine vision system need and what options are available for those needs. One application was to replacing people in repetitive quality assurance tasks in packaging. The machine vision system should be able to detect seam thickness and quality, location of a product inside of packaging, and dirt and dust particles inside packaging. The system should also be able to match the machine speed to not slow down the production, and handle different sizes of products (between 5×5cm, 15×15cm, and 11×20cm). The research gave the requirements to 4 unnamed companies to fill using rejected products as samples. Important findings the companies stated: placement of camera, placement of product, size of product, product’s contrast to background, and lighting play a large role how the machine vision system would succeed. Notes on how to achieve higher success rate were: background should be text free and have high contrast difference to the product, different parts of the product that should be measured separately should have high contrast difference, multiple pictures from different angles and using different lighting, and correct camera. The end result was that additional research has to be done to fill all requirements since packaging material position detection had untold issues. (Halinen, 2017)

“*Intelligent fish feeding based on machine vision: a review*” article has collected knowledge from multiple articles about utilizing machine vision to monitor fish growth and amount and

to identify fish species and individual fish. Issues this study found that affected directly the machine vision model sturggled to identify individuals when using them as targets but this could be partially solved by used the whole group as target object. The machine vision model had varying speed and accuracy in processing due to background and difference between fish species which cause different amount of layers in a deep network structure. A proposed solution was invetigate maximum time available and optimizing accuracy to that time. As an addition, deep learning models are significantly slower than traditional image processing models but traditional image processing models couldn't handle complex images thus model should be selected via application needs. One problem for the model was how single angle images lack information. A solution was presented with multi-angle camera system that constructed 3D model from the fish to be analysed. (Zhang *et al.*, 2023)

(TODO: comparison)

## 2.2 Machine learning

This sub-chapter will present how other have taught their machine vision model and how that compares to this research.

The book "*Machine Learning*" published 2021 defines machine learning as a technique to improve system performance using computational methods to learn from experience (Zhou, 2021, p. 2). Another book also called "*Machine Learning*" published 2026 refers machine learning as "*data-adaptive*" models (Theodoridis, 2026, p. 2). Alas, this study defines machine learning as the process of algorithmically generating models that approximates by teached data.

The 2026's "*Machine Learning"* book lists four technical challenges in machine learning. The first is creating a system than constantly learn while interacting with changing enviroment and while performing decisions securely, timely, and robustly. The second is creating a system that do not compromise privacy and security of users while enabling personalization of applications and services. The third is secure use of confidential training data. The fourth is developing systems to address the calculation need of future applications of

machine learning. The book also states social challenges from explotation of the data used by these applications. (Theodoridis, 2026, p. 8)

"*Intelligent fish feeding based on machine vision: a review*" article has collected issues about machine learning. They had several problems with the training data that affected machine learning such as dificulty to obtain data from multiple sources due to lack of manpower and funds, data quality was uneven due to natural changes in angles, targets, and their back-ground, and time-consuming task of creating data sets for proper teching process. (Zhang *et al.*, 2023)

The 2021 published book *"machine learning"* brought up a issue called "*inductive bias*". It is about how machine learning handles the issue of how to predict a outcome from set that does not exist in teaching data set. Each time a machine learning algorithm generates a model, that model may predict the outcome differently. (Zhou, 2021, p. 7-11)

(TODO: comparison)

## 2.3 Camera systems

This sub-chapter will present how other have chosen and implemented their camera systems and how that compares to this research.

(TODO: definition?) This research defines camera systems as the mechanical sensors that are read for image data. Stereo vision is to capture image data with two cameras (*OpenCV: Epipolar Geometry*, no date).

A study called *"3D Dace Tracking Using Stereo Camera"* researches how using multiple cameras and forming a 3d image from the data can improve machine vision. The issues this research presents are the difficulty of tracking human faces due to their capacity to change with emotion, communication, and physiology, image resolution's impact, illumination variations and object's self caused occlusion. The study refers to another study that states 3d trackers are comercially available (Roy-Chowdhury and Xu, 2014) but they are usually not suitable for motion tracking. 3D face tracking image processing with ligthing from multiple angels enables a higher quality 3D image which partially fixing self-occlusion problem. Low

resolution causes problems in any tracking or detection algorithm while syper resolution requires precise modeling of the algorithms to track facial changes but when done correctly and feeding back previous frame for tracing the next frame, the tracking method improved significantly. As challenges for stereo vision cameras, facial feature detection being robust has been subject where strategies were hard to develop, wider separation of the cameras give more accurate 3D model but makes connecting the pictures harder, stereo vision requires additional cameras which are expensive for additional computational input, occlusion caused by postion of the cameras which can be fixed with placement of cameras and additional cameras, and the depth resolution is affected by the correlation algorithm and resulation of images. The study suggest, as future work, to look into changing enviroment, occlusions, expressions, noise reduction, and facial resolution. (Alqahtani, Chandran and Banks, 2018)

A study called *"Three-Dimensional Head Pose Estimation Using a Stereo Camera Arrangement"* is a continuation to the reseach *"3D Dace Tracking Using Sterao Camera"* which is about how using 3D camera system improves machine vision (Alqahtani, Chandran and Banks, 2018). To improve the framework, this reseach utilizes perspective-n-point problem, which is a matematical way to estimate the pose of a camera from 3D world coordinates that are projected to 2D plane. This then creates an instance of perspective-n-point problem which when solved estimates the camera position. This study's justification for this issue is that 2D data has issues with resolution and 2D head pose estiomation data from single image is scarce which is compensated with multiple images or video. 3D scanning could solve the issues with 2D but it has its own issues which cause it to not be consumer friendly. The study notes that depth cameras are more accessable but lack in quality and precision. The study made and experiment that used 30 videos with each 300 frames and gained 98.6 as a result. (AlQahtani *et al.*, 2018).

A study called *"A novel Stereo Vision Sensor for Fast Moving Objects"* researches how a line scan camera, frame camera, and a line lasers can create a system to make 2D scans and 3D reconstructions simultainously from fast moving objects. The study makes a physical experiment showing that the system works robustly and accurately. The system works by first the line scan camera scans the moving object while the light stripe images are obtained by the frame camera. Then a epipolar line which is a geometric constraint used to find corresponding points between two cameras, is calculated for each pixel in the line scan camera enabling more effective searching (*OpenCV: Epipolar Geometry*, no date). Then using

binocular stereo vision sensor model, 3D coordinates are calculated and then x coordinate is calculated from the capture interval and the time of capture. The experimental system this research created achieved 0.28 mm calculated accuracy via $RMSE_M$ method and was further evaluated with reconstruction of workpieces from scans which matched with the actual shapes. The research notes in its conclusion that aligning the line scna camera and line lasers can improve the accuracy. (Wu *et al.*, 2018)

(TODO: comparison)

# 3 Methodology

This chapter covers the research methodology used in this thesis, why it was chosen, and what steps it requires and how they are taken.

## 3.1 Choosing Methodology

This thesis is required to compare existing machine vision models and camera systems, select a most fitting for a bitwell, design a usable framework to implement a trained machine vision model, train the model, and create a working system as proof of concept while documenting anything done in this thesis from reasoning and choises to physical implementation.

This research uses *"Guiding the selection of research methodology in industry–academia collaboration in software engineering"* as guide to choose what methodology should be used (Wohlin and Runeson, 2021). This guide was chosen due to its focus on methodologies which are used a lot in software engineering and follow pattern "describe-solve-practice" as stated on the guide. "Descripe" refers to describing the problem, "solve" refers to developing a solution to the problem, and "practice" refers to evaluating the solution how well it address the problem. These three methodologies are: Design Science, Action Research, and Technology Transfer Model. A fourth one called Design Research Methodology is also mentioned in Appendix A but according to the guide it just complements the other three. (Wohlin and Runeson, 2021)

Design Science has focus on the artifact made from a design and the documentation from the design process. Action research has its origins in social science and has focus on making a change and studying its consequences in iterative cycles. Technology transfer model has focus on industry challenges and how to academically solve them. It includes tight collaboration between researchers, champion, and practitioner. It isn't a methodology but a model which could be used as methodology since it has clear steps to conduct the research and the study proposes reformulating it into a research methodology. (Wohlin and Runeson, 2021)

This master's thesis will choose Design Science as its main methodology but enhances it with proposed Technology Transfer Model. This is due to the fact that as the guide states, Technology Transfer Model isn't a methodology (Wohlin and Runeson, 2021) but this thesis will benefit from the activities proposed in the model since this thesis will be done with and for bitwell. Design Science was otherwise clear choise as the intent was to create a new machine vision system from existing knowledge thus not changing the old system which eliminates Action Research as a methodoly choise.

## 3.2 Design Science

Design Science focuses to the object of the study called *"artifact"* and has to major activities designing and evaluating (the book calls it investigating) according to a book called *"Design Science Methodology"* (Wieringa, 2014, p. 3). Design Science has four existing (problem identification, solution design, evaluation and summarise results) research phases according to (Offermann *et al.*, 2009). They consist of 11 activities: Identify problem, Literature research part 1, expert interviews, pre-evaluate relevance, design artifact, literature research part 2, refine hypothesis, expert survey, laboratory experiment, case study/action reseach and summarise results (Offermann *et al.*, 2009). Notable here that not all these activities are done in each Design Science research but are mere suggestions what could be done. These activities can be encapsulated into three categories: Problem identification, solution, and evaluation (Offermann *et al.*, 2009; Wohlin and Runeson, 2021). Here evaluation includes summarise results as an activity. For comparison, an article called *"A Design Science Research Methodology for Information Systems Research"* places 6 steps for Desing Science: Problem identification and motivation, definition of the objectives for a solution, design and development, demonstration, evaluation, and communication (Peffers *et al.*, 2007).

The definition of an artifact is vague (Wieringa, 2014; Wohlin and Runeson, 2021) but designing it and evaluating it thus making Design Science research contributes in terms of knowledge by answering knowledge questions (Wieringa, 2014, p. 5-6) contributing with "technological rules" (Aken, 2004), the knowledge in the context of the project (Wieringa, 2014, p. 3), and the practice while being done in the context of the practice (Wohlin and Runeson, 2021). Design sciences another contribution to design problems with the produced

artifact which makes improvement by interacting in the context of a problem (Wieringa, 2014, p. 3-6).

(ToDo/ask?: explain each activity and phase?)

## 3.3 Technology Transfer Model

Technology transfer model main point is to ensure that the developed artifact is usable in the industry side. This is achieved with collaboration between the industry and the academia thus the solution is validate multiple times through out the creation process. The activities presented in study *"A Model for Technology Transfer in Practice"* to technology transfer model are Indrustrial challenge, Problem statement and state-of-the-art, candidate solution, academic validation, static validation, dynamic validation, and release solution (Gorschek *et al.*, 2006). The guide *"Guiding the selection of research methodology in industry–academia collaboration in software engineering"* suggest reformation to six activities for it become more research methodology like and naming it Technology Transfer Research Methodology. The six suggestes activities are: Identify the industrial challenge, assess practice and formulate a research objective, study state-of-the-art, develop one or more candidate solution(s), evaluate the solution(s) (academic, static, and pilot), and move the chosen solution into practice and evaluate. (Wohlin and Runeson, 2021)

(Todo/ask?: open up activities?)

## 3.4 Methodology Practice

The book *"Design Science Methodology"* talks about research goals and how to categorise them and according to the descriptions, this thesis goal is *"curiosity-driven"* artifact design goal (Wieringa, 2014, p. 13-15). This means this thesis aims to answer knowledge questions while designing an artifact to improve in the context of problem. To reach these goals, this research will be done in a following way mimicking what descriped in chapters 3.2 and 3.3.

The structure of this thesis will follow selected activites from what (Offermann *et al.*, 2009) had presented. These activities are identify problem, literature research part 1, pre-evaluate

relevance, literature research part 2, design artifact, refine hypothesis, laboratory experiment, and summarise results. These activiteis will be used with technology transfer model's steps to modify them to acknowledge industry's needs more.

Design Science offers following steps. In problem identification, the problem is identified and its relevance is figured out. In this thesis, it is in introduction and this step is done for this thesis in collaboration with Bitwell as technology transfer research methodology suggests (Wohlin and Runeson, 2021; *Bitwell Oy*, no date). In literature research part 1, with technology transfer research methodology study state-of-the-art (Wohlin and Runeson, 2021), documented as related research, the relevance and relation to academic research and potential issues and obstacles to the solution are identified. In pre-evaluation, documented in introduction, the problem is evaluated and hypothesises are created. The form of hypothesis should be: if problem is solved with the solution, observable reaction will happen which supports a cause. In literature research part 2, document as parts of chapters 4,5, and 6, additional information for design phase is gathered. In design artifact, with technology transfer research methodology develop cadidate solution(s) (Wohlin and Runeson, 2021), which are documented partially in chapters 4, 5, 6, and 7, the artifact is designed and created to solve the identified problem and design choices are documented. In the refine hypothesis, documented in introduction, the hypothesises are made smaller and more precise for the research. In the laboratory experiment also called academic validation in technology transfer research methodology (Wohlin and Runeson, 2021), documenter in artifact chapter, the artifact is evaluated in controlled environment multiple time to gain statistical data for evaluation. As additional evaluation from technology transfer research methodology, static and dynamic validations are done. In static evaluation, the solution is discussed with the Bitwell and feedback is collected (Wohlin and Runeson, 2021). In dynamic validation, the artifact is used in suitable situation as close as possible of an actual setting (Wohlin and Runeson, 2021). Static and dynamic validation are stored in chapter 7. In the summarise results the whole research process and its results are compressed in abstract and conclusion chapters. (Offermann *et al.*, 2009)

# 4 Machine vision models

In this chapter, this research explains basics of machine vision, what models are available, how they compare to each other, what model this research ends up using, and reasoning why that model was selected.

## 4.1 How machine vision works

According to Halinen's master thesis, machine vision system consist of four parts: forming of image, capturing of image, image processing, and control system (Halinen, 2017).

"*Intelligent fish feeding based on machine vision: a review*" article has collected knowledge from multiple articles about utilizing machine vision to monitor fish growth and amount. It explains how machine vision itself works and divides it into two parts: image processing and analysis. Image processing is about filtering the relevant information from the image. This could be done with image denoising, image enhancement, image segmentation, image binarisation, morphological processing, and contour extraction. Image analysis is about collecting the information in the images. Image analysis could be done with image regression, image classification, and image clustering. The review divides the whole machine vision process into traditional image processing and analysis and into deep learning. This division depends which techniques are used. In traditional method, image is processed and then analyzed. (Zhang *et al.*, 2023)

In image processing target is usually separated from background in image segmentation process which improves image analysis accuracy when done correctly. Image segmentation can be done with threshold-based method, edge detection based method, and specific theory-based method which itself can be divided into wavelet analysis, genetic algorithm, and active contour. When image segmentation is used to segment a fish, it has a issue that same image may have overlapping fish. To fix this issue, there are three methods: the watershed method, ellipse fitting method, and the segmentation point detection method. The review focused on the point detection method where convex point detection, the shortest centroid distance, and the segmentation lines were achieved via corner detection and non-maxium distance to

segment the overlapping fish. This point detection method fixed the issue of overlapping fish. (Zhang *et al.*, 2023).

Image feature extraction is focusing on the whole picture like how many fish are in the image. Methods for image feature extraction include shape features, texture features, space features, and color features. Shape features are information from the shape of objects. Texture features are information collected from textures. Space features are information from the distance between objects and the objects positions. Color features are information from the objects colors. (Zhang *et al.*, 2023)

Image classification, feature fitting, and regression are done via machine learning algorithms with help of information gained from the image feature extraction. In this study, the goal of these methods is to determine fish mass and behavior state. The machine learning types usually used in with small sample data are back propagation neural network and support vector machine. (Zhang *et al.*, 2023)

In deep learning is about taking a large amount of samples and use those to teach a neural network to do image recognition and analysis on its own. It could identify subtle features such as fish illumination, translation, and rotation. (Zhang *et al.*, 2023)

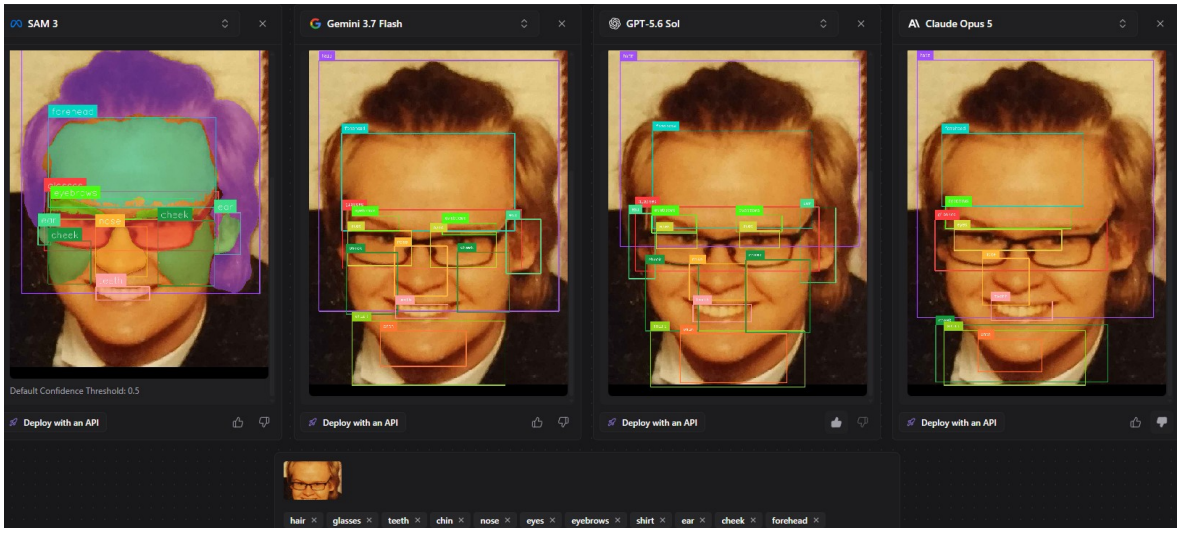
## 4.2 Available models

Geeksforgeeks website presents their list of top machine vision models and the 6 categories they belong: Convolutional Neural Networks (CNN), Region-based Convolutional Neural Networks (R-CNNs), You only look once (YOLO), Vision Transformers (ViTs), EfficientNet, and Contrastive Language-Image Pretaining (CLIP). The top models are: VGGNet, GoogLeNet (Inception), ResNet, AlexNet, MobileNet, Fast/Faster/Mask R-CNN, YOLOs V3-V5, Swim Transformer, Data-efficient Image Transformer, Pyramid ViT, EfficientNet-B0 to B7, EfficientNetV2, EfficientNet-Lite, CLIP, and OpenClip. (*Top Computer Vision Models*, 2025)

Top models according to a blog on Saiwa.ai website are: YOLOv11, ViT, Segment Anything Model (SAM) 2, Mobile ViT / Efficient ViT, ConvNeXt / ConvNeXtV2, U-Net, Detection Transformer, RetinaNet, OpenSeeD (Open-Vocabulary Segmentation), and DenseNet. (Komeili, 2023)

Top models according to Shaik Hamzah Shareef on site analyticsvidhya are: AlexNet, VGG-16 and VGG-19, ResNet, DenseNet, EfficientNet, ConvNeXt, R-CNN and its variations Fast/Faster/Mask, YOLOs V1-V12, Single Shot MultiBox Detector, U-Net, Detectron2, DINO, CLIP, BLIP, ViTs and their successors, and SAMs. (Shareef, 2025)

Roboflow is a website evaluates and ranks machine vision models. It has six different categories to each evaluation done and an overall ranking. Any user can even compare different models on any of the tasks for free (Fig X). Top 10 models according to roboflow website's overall ranking: SAM 3, Gemini versions 3 flash, 2.5 pro, 3.5 flash, and 2.5 flash, Qwen 3.8 Max, GPT 5.5 and 5.4, Kimi K3, and GPT-5 Mini. (*Vision AI Models: Explore & Try the Latest | Roboflow Playground*, no date)




(fig x. Comparison of Machine vision performance on object detection of a picture of a handsome fellow. (*Vision AI Models: Explore & Try the Latest | Roboflow Playground*, no date))

Top models according to site Encord are categorized into Image classification, Object detection, and Image segmentation (sematic, instance, and panoptic). Top Image Classification are: Contrastive Captioner (CoCa), Pathways Language and Image (PaLI), CoAtNet-7, DaViT, and FixEfficientNet. Top object detection are Co-Detr, InternImage, Focal-Stable-DINO, EVA, and YOLOv7. Top sematic image segmentation models are ONE-PEACE and Mask2Former. Top instance segmentation are Mask Frozen-DETR and DiffusionInst-SwinL. Top panoptic segmentation are PanOptic SegFormer, and K-net. (Sajid, 2025)

Top models according to site unitlab are: YOLO, DETR, Faster R-CNN, Mask R-CNN, and EfficientDet. (Barotov, 2025)

Previous sources had many overlapping rankings of AI models but most mentioned were YOLO (5 sources out of 6), Faster R-CNN (3 sources out of 6), Mask R-CNN (3 sources out of 6), and SAM (3 sources out of 6). Hence this master's thesis will focus on Researching YOLO models and if they do not provide adequate results, next in research are faster R-CNN, then Mask R-CNN, and then SAM.

### 4.2.1 YOLO

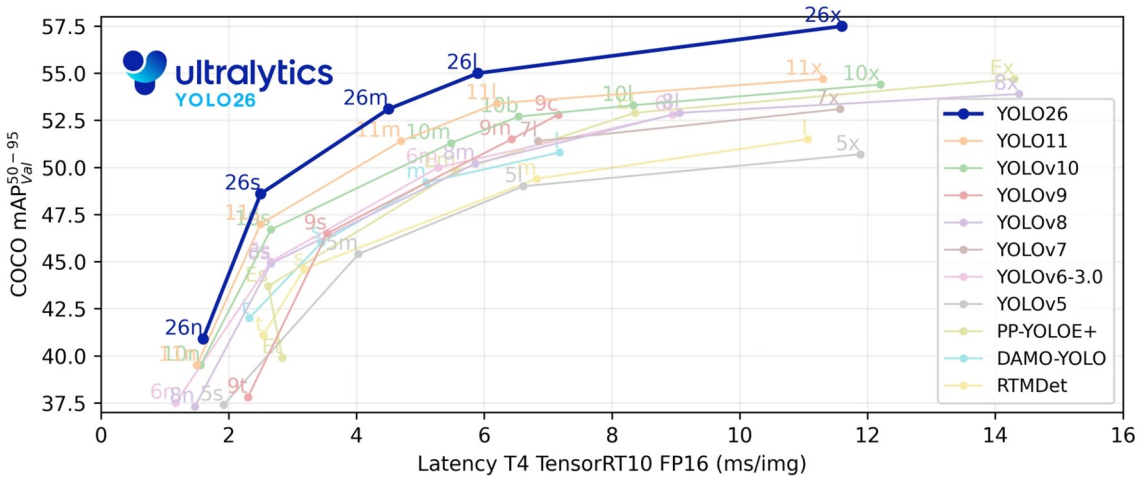
YOLO is a machine vision algorithm which first version was published 2015 (Hussain, 2023; Sapkota *et al.*, 2025). There are YOLO's main versions from v1 to v10, 11, 12, and then YOLO26 as the newest at the time of writing august of 2026 (Boesch, 2026). According to the ultralytics website, YOLO2027 is going to be published (*YOLO Object Detection & Segmentation*, 2023). YOLO's fast development can be contributed to the fact that there is no copyright or patent restrictions (Hussain, 2023).

YOLO is an one-stage detector. Usually object detection is classified into two stages: Feature extraction (first stage) and regression and classification (second stage). Systems called two-stage detectors use these separately can achieve high accuracy with a cost of a high computational demand while systems called one-stage detectors merge these stages into one reducing the computational demand (Hussain, 2023; Sapkota *et al.*, 2025). YOLO's lightweight, accuracy, and edge-friendly deployment conditions has made it highly compatable to the industrial requirements. (Hussain, 2023)

YOLO has been development throughout the years. Here is a list what each version main changes/features are accoding to studies (Hussain, 2023, 2024; Terven, Córdova-Esparza and Romero-González, 2023; Sapkota *et al.*, 2025; Boesch, 2026), and to the ultralytics website (*YOLO Object Detection & Segmentation*, 2023):

- YOLOv1: Used simplified convoluted neural network (CNN) as backbone and made basic predictions for bounding boxes.
- YOLOv2: Changed backbone to DarkNet-19 and added batch normalization, direct location prediction, and improved anchor boxes with K-means clustering.
- YOLOv3: Changed backbone to DarkNet-53 and added three-scale detection and residual connections.
- YOLOv4: Changed backbone to CSPDarkNet-53 and added Path Aggregation Network (PANet).
- YOLOv5: Changed backbone to CSPNet and improved on dynamic anchor refinement and improved Grid sensitivy.
- YOLOv6: changed backbone to EfficientRep and added RepVGG/CSPStackRep (depending on variant) block to backbone and improved on PANet, added self-distillation strategy, and optimized with quantization.
- YOLOv7: Used EfficientRep and added dynamic label assignment with coarse and fine labels and batch normalization in conv-bn-activation.
- YOLOv8: Improved Path Aggregation Network with Dynamic Kernel Attention
- YOLOv9: Added multilevel auxiliary feature extraction
- YOLOv10: Uses enhanced version of CSPNet as backbone and optimized system
- YOLO11: Used C3k2 backbone and used C2pSA for spatial attention
- YOLO12: Shifted toward attention-centric design with addition of Area Attention and Residual Efficient Layer Aggregation Networks (R-ELAN)
- YOLO2026: Has optional one-to-one detection without non-maxium suppression (NMS), simplifying deployment and reducing post-processing.
- YOLO2027 (not released at the time of writing): Added dual-scale detection, stronger small-object detection, and foreground alignment supervision to extract an object from background.

YOLO2026 seemed most promising to the given requirements. Its accuracy compared to other models (fig x.) and easy deployment while being lightweight makes it a valid solution for the machine vision system. Next is the training/fine-tuning of this model.




(fig x. Ultralytics accuracy-latency graph (Boesch, 2026))

# 5 Training of the machine vision model

This chapter will go over how machine vision model can be trained and how this research will train it. Training related data will be displayed which includes method used, the cost of training, and details of training data such as format and amount.

## 5.1 Requirements for training

## 5.2 Training data

"*Intelligent fish feeding based on machine vision: a review*" article focuses on utilizing machine vision to monitor fish growth and amount. The research compares different types of training data: Public data, internet data, and experimental collection data. Public data is easily accessible and is a large collection but has a lot of differences between pictures like different environment or different quality of picture. It is stated that they are designed to fish species identification which focuses on differences between fishes species not in individual differences like mass or growth progress. Internet data has large volume of data, rich diversity in data, and easy to access but some of the date is irrelevant to the training and there is a lot of inconsistencies between images such as some images are photos and other are drawings. Experimental collection data has to be collected by the researchers themselves thus giving them control what the data consists but requires effort. (Zhang *et al.*, 2023).

## 5.3 Training on hardware or on cloud service

There are currently two ways to gain access to computational power to train AI: buying hardware or buying cloud computing from cloud provider (source?). Cloud provider is a business that sells cloud infractucture such as vitual machine, servers, databases, and hosting (*Cloud service providers: Top clouds and how to choose your cloud provider*, no date).

This thesis will compare top three companies for cloud services which are Amazon, Microsoft, and Google (Saraswat and Tripathi, 2020; Dubendorfer, 2025; *Cloud service providers: Top clouds and how to choose your cloud provider*, no date) and multiple others found by searching “cloud computing service”.

### 5.3.1 Comparison Research on Cloud Providers

Here’s a table (Fig Unknown.) about different cloud services offering calculation power for AI training. The websites themselves rarely note the speed of their RAM, VRAM, or GPUs. Note: Amazon Web Service (AWS) Sagemaker AI, Microsoft Azure, and google cloud platform has regional prices and this thesis uses stockholm server for AWS, East US 2 servers for Microsoft Azure, and cheapest available for Google cloud platform prices. (How to fit the table on this thesis?: Annouce its existance, publish and link but no actual images or tables here. Data was collected on 19.8.2026)

### 5.3.2 Comparison Literature on Cloud Providers

A website called DigitalOcean has provided a neat list how to choose a cloud provider. The first item is pricing. The website tells to keep a eye on different costs as providers like AWS has a complex pricing which might cause a lot more costs than initially thought. The second item is product offering. Decide what product you need and seek it. The third is support and it is usually overlooked. Many cloud providers do not offer support for free. The fourth item is data center locations since they may not have a data center in desired locations. The fifth item is documentation which is essential for solving problems that might arise when using a cloud providers. The site recommends looking into the documentation, community forums, and tutorials. The last item is supported operating systems as some applications require specific operating system. (*Cloud service providers: Top clouds and how to choose your cloud provider*, no date)

Of the items from previous pharagraph, this master’s thesis prioritizes cost, product offering, support, and documentation in order they were listed. Data center location and supported

operating systems are ignored as it should not affect the artifact and ideally the artifact should be able to be run in any major operating system.

A website called varionis has a blog by Dubie Dubendorfer about comparing AWS, Azure, and Google as cloud servicer providers.

https://www.varonis.com/blog/aws-vs-azure-vs-google

https://search.brave.com/search?q=how+to+compare+cloud+providers&conversation=0976fc390651f52162ed84ff1bf721b50f37

#### 5.3.3 Own hardware.

#### 5.3.4 Training method choice.

### 5.4 Training a prototype.

# 6 Camera systems

This chapter goes over how different camera systems compare and how this research chose what it uses in implementation.

Sony: IMX 4K, 1/1.8 inch, 20x / 40x optic zoom, 256 encoder,  2

# 7 The artifact

This chapter tells how the whole system was put together, how different parts communicate through interfaces, and how the trained machine vision model was run.

# 8 Discussion

This chapter is a reflect how the research went, how well this answered the research questions, how this research compares to related research, and more insight to how to improve the system.

# 9 Conclusions

# 10 Future work

This chapter displays what could be possible research questions to the next research and why they are significant.

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# Appendix